Biz. Solutions

Platform Basics

PowerApps 101:

From Curious to Expert

Low Barrier

Charles Lakes II

Microsoft® MVP Most Valuable Professional

About Me:

- Pro-Code Developer
- Low-Code Maker
- Biz. Solution Consultant
- Middle-child

Agenda:



Part #1:
East Blue



Part #2:
Enies Lobby



Part #3:
Thriller Bark



Part #4:
Whole Cake



Part #5:
Wano



Part #6:
Egghead

Schedule:

Part #1 | 9:00 AM - 10:00 AM	Break | 10:00 AM - 10:10 AM	Part #2 | 10:10 AM - 11:10 AM
Break | 11:10 AM - 11:20 AM	Part #3 | 11:20 - 12:30 PM	Lunch | 12:30 PM - 1:30 PM
Part #4 | 1:30 PM - 2:30 PM	Break | 2:30 PM - 2:40 PM	Part #5 | 2:40 PM - 3:50 PM
Break | 3:50 PM - 4:00 PM	Part #6 | 4:00 PM - 5:00 PM	

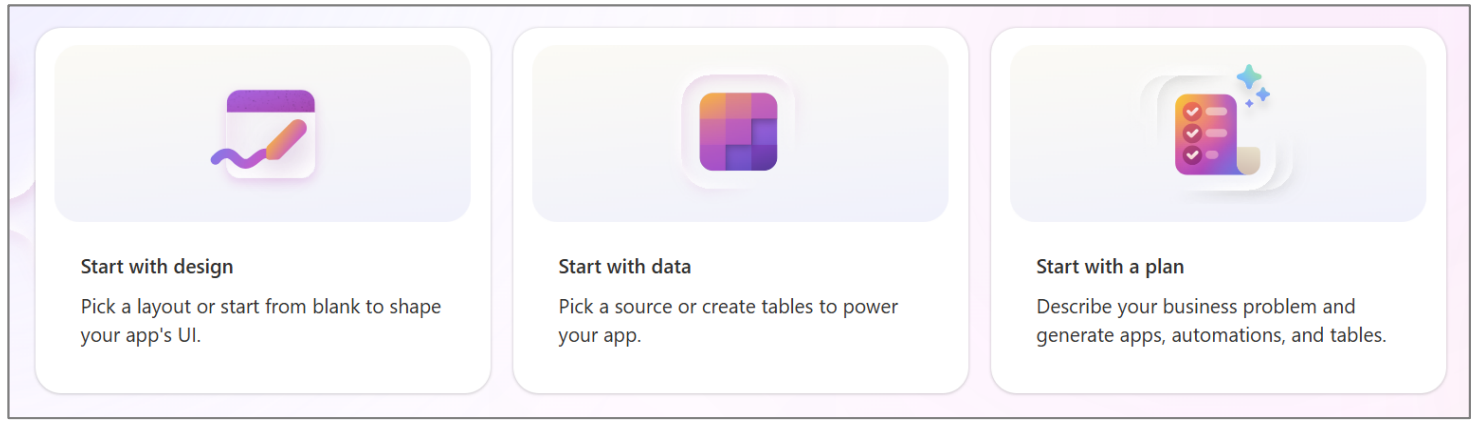
Power Apps:

Canvas apps are custom business applications you can create in Microsoft Power Apps without writing code. Think of a canvas as a blank design surface where you can drag and drop components to build exactly the user interface you need. Whether you're creating a simple data entry form or a complex business workflow, canvas apps give you the flexibility to design solutions that work the way your business works.

With canvas apps, you can:

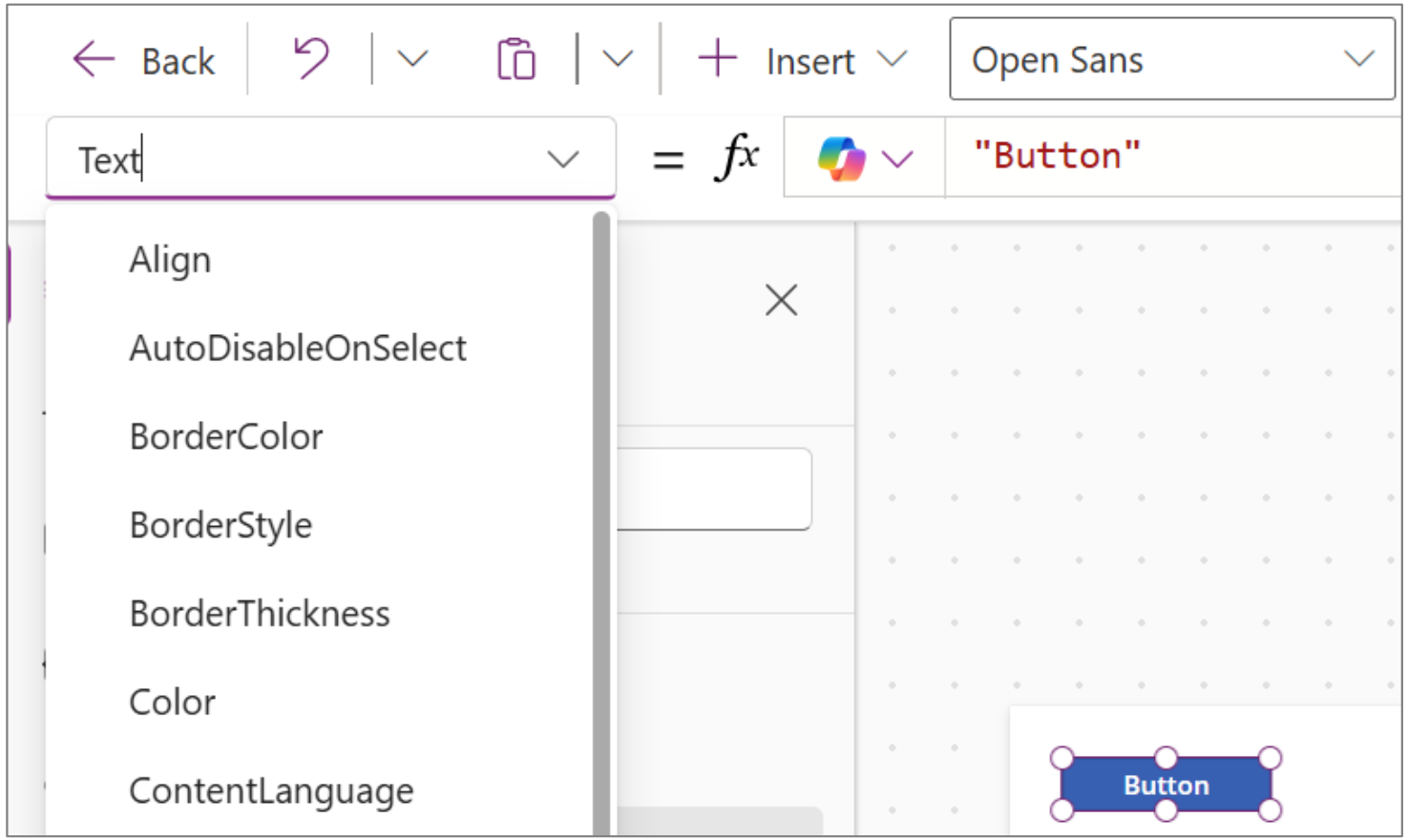
- Connect to data from hundreds of sources, including Microsoft 365, Dataverse, SharePoint, and other data sources
- Create responsive designs that work seamlessly across browsers and mobile devices
- Use AI-powered Copilot to build apps through natural language conversations
- Share your apps securely within your organization
- Embed apps directly in SharePoint, Power BI, and Microsoft Teams

Part #1: East Blue



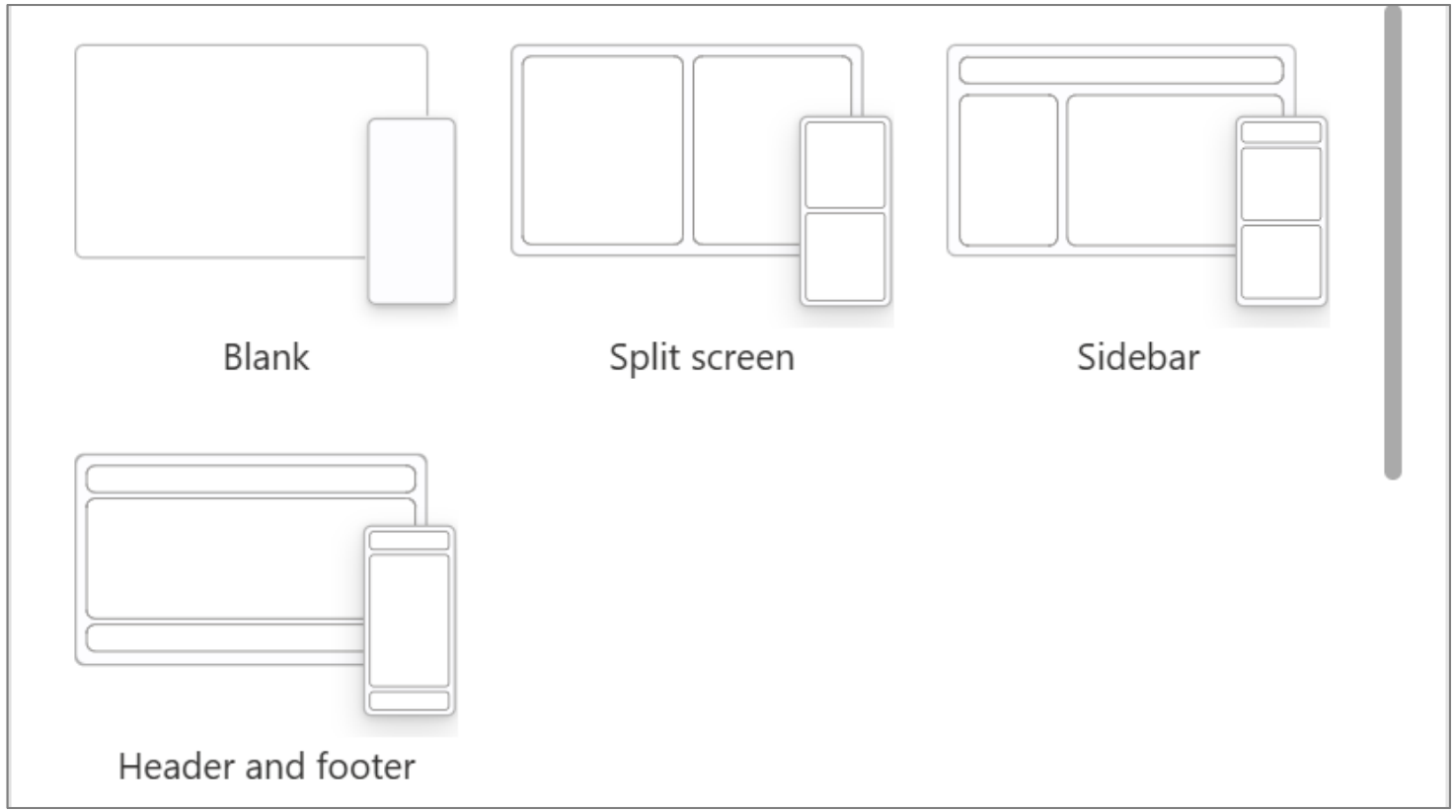
- ❑ App Types
  - Canvas
  - Code
  - Model-Driven
- ❑ Control Types
  - Formula Bar
  - Properties
- ❑ Screen Types
  - OnStart
  - StartScreen

Part #1: East Blue



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Part #1: East Blue



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Part #2: Enies Lobby

Data type	Description	Examples
Boolean	A *true* or *false* value. You can use it directly in **If**, **Filter**, and other functions without a comparison.	*true*
Choice	A choice from a set of options, backed by a number. This data type combines a localizable text label with a numeric value. The label appears in the app, and the numeric value is stored and used for comparisons. The **Type** function supports this data type if you use an instance of a **Choice** field by name.	**ThisItem.OrderStatus**
Color	A color specification that includes an alpha channel.	**Color.Red** **ColorValue("#102030")** **RGBA(255, 128, 0, 0.5)**
Currency	A currency value stored in a floating	123

- Power Fx
  - Data Types
  - Formulas
  - Global Variables
  - Context Variables
  - Collections
- Data Sources
  - Excel
  - SharePoint
  - SQL Server
  - Dataverse
- Copilot

Part #2: Enies Lobby

Formulas in canvas apps combine several elements. Here are the main types:

- **Functions** take parameters, perform an operation, and return a value. For example, **Sqrt(25)** returns **5**. Functions are similar to Microsoft Excel functions. Some functions can have side effects, like **SubmitForm**, which you use only in a behavior formula such as **Button.OnSelect**.
- **Signals** return information about the environment. For example, **Location** returns the device's current GPS coordinates. Signals don't take parameters or have side effects.
- **Enumerations** return a predefined constant value. For example, **Color** is an enumeration with predefined values like **Color.Red** and **Color.Blue**. Common enumerations are included here, and function-specific enumerations are described with the function.
- **Named operators**, like **ThisItem** and **Self**, let you access information from within a container.

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Part #2: Enies Lobby

Global variables cannot use the same name as an existing collection or control. It can use the same name as a context variable. To disambiguate between the two, use the disambiguation operator.

Set has no return value, and you can use it only within a behavior formula.

Syntax

Set(*VariableName*, *Value*)

- *VariableName* - Required. The name of a global variable to create or update.
- *Value* - Required. The value to assign to the context variable.

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Part #2: Enies Lobby

Every context variable is scoped to a screen. If you want to define a context variable on one screen and modify that variable from another screen, you must build a formula that's based on the **Navigate** function. Or use a global variable.

UpdateContext has no return value, and you can use it only within a behavior formula.

Syntax

UpdateContext({ *ContextVariable1*: *Value1* [, *ContextVariable2*: *Value2* [, ...]] })

- *ContextVariable1* - Required. The name of a context variable to create or update.
- *Value1* - Required. The value to assign to the context variable.
- *ContextVariable2*: *Value2*, ... - Optional. Additional context variables to create or update and their values.

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Part #2: Enies Lobby

Formula	Description	Result
ClearCollect(IceCream, { Flavor: "Strawberry", Quantity: 300 })	Clears all data from the **IceCream** collection and then adds a record that includes a quantity of strawberry ice cream.	Flavor | Quantity "Strawberry" | 300 The **IceCream** collection has also been modified.
Collect(IceCream, { Flavor: "Pistachio", Quantity: 40 }, { Flavor: "Orange", Quantity: 200 })	Adds two records to the **IceCream** collection that includes a quantity of pistachio and orange ice cream.	Flavor | Quantity "Strawberry" | 300 "Pistachio" | 40 "Orange" | 200 The **IceCream** collection has also been modified.

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Part #2: Enies Lobby

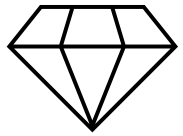
Types of connectors

We offer a wide variety of ready-made (prebuilt) connectors, but sometimes you might want to call APIs, services, and systems that aren't available as prebuilt connectors. To accommodate different needs, you can choose your connector type.

There are two types of connectors:

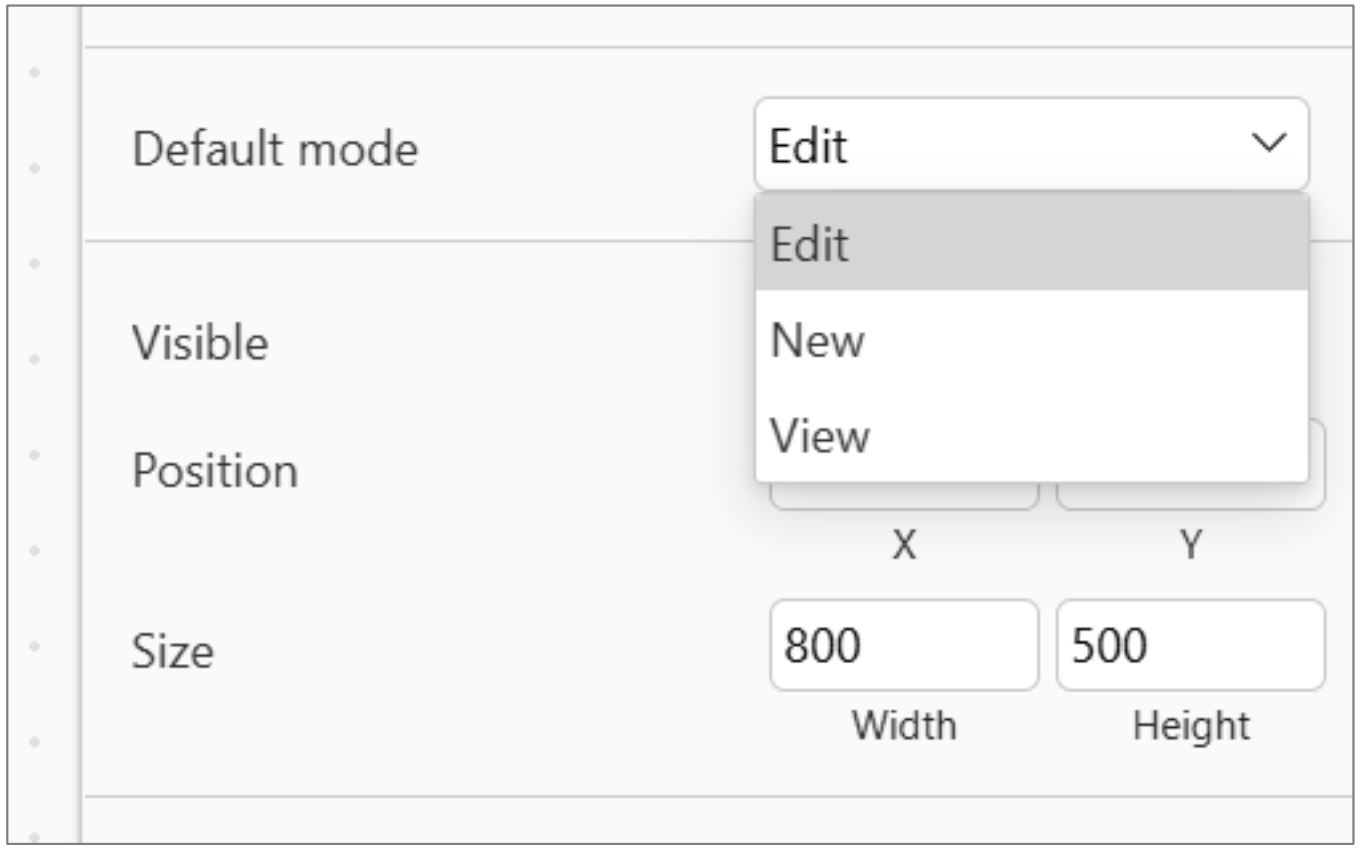
- **Prebuilt**: Predefined connectors that can be used without you making modifications. They address automating commonly used business process scenarios.
- **Custom**: Function-based connectors that you build to support your tailored scenarios.

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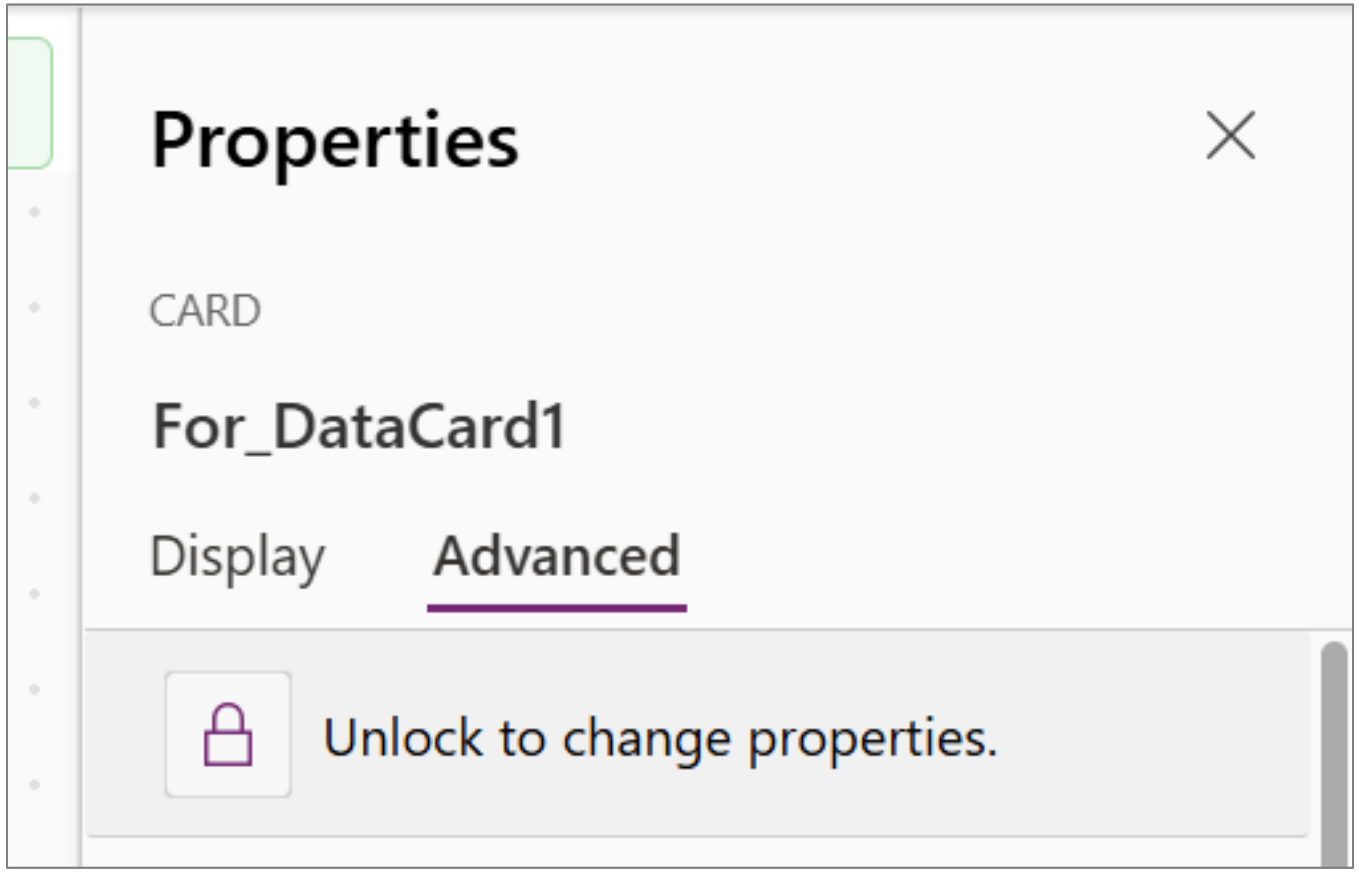
- ❑ Copilot

Part #3: Thriller Bark



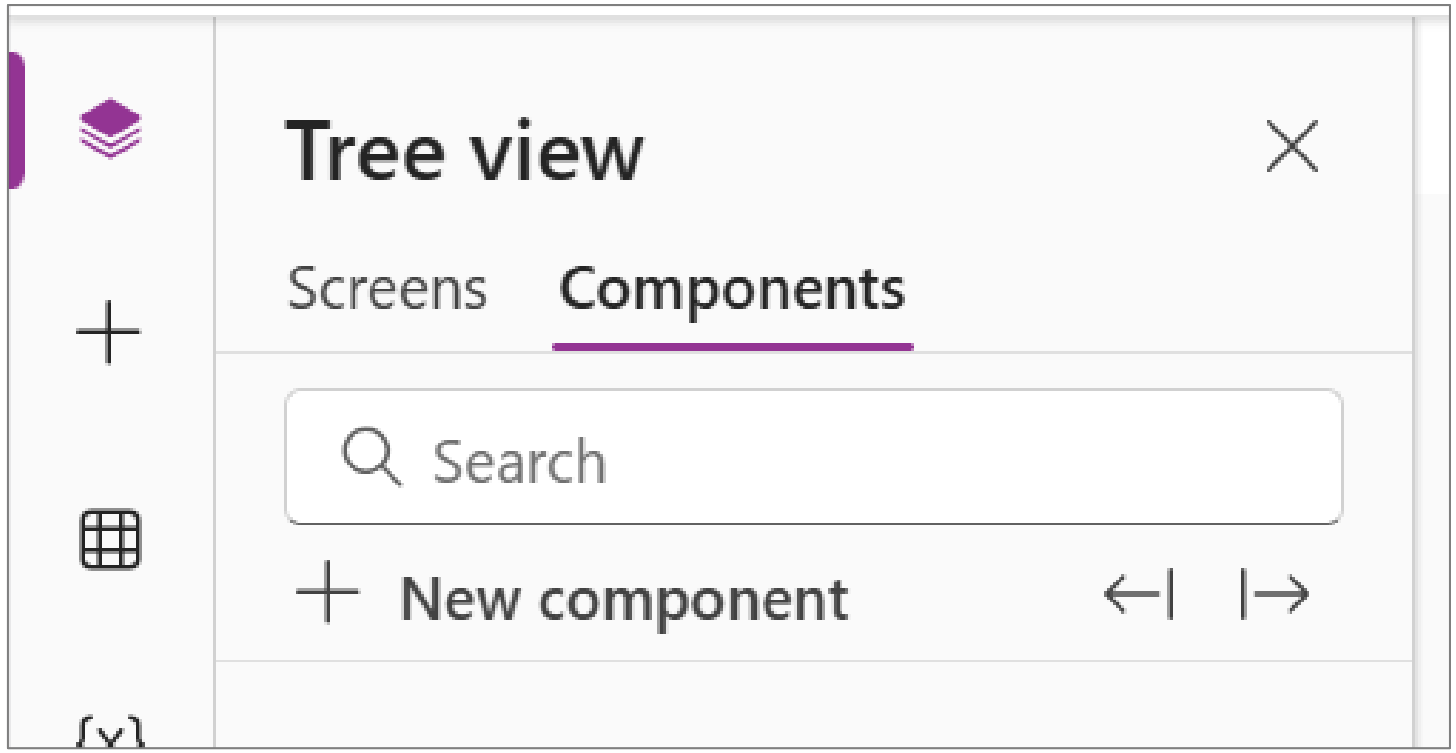
- Data Forms
  - Default mode
  - Unlock cards
- Modularity
  - Components
  - Containers
- Advanced Power Fx
  - With() Expressions
  - User Defined Functions

Part #3: Thriller Bark



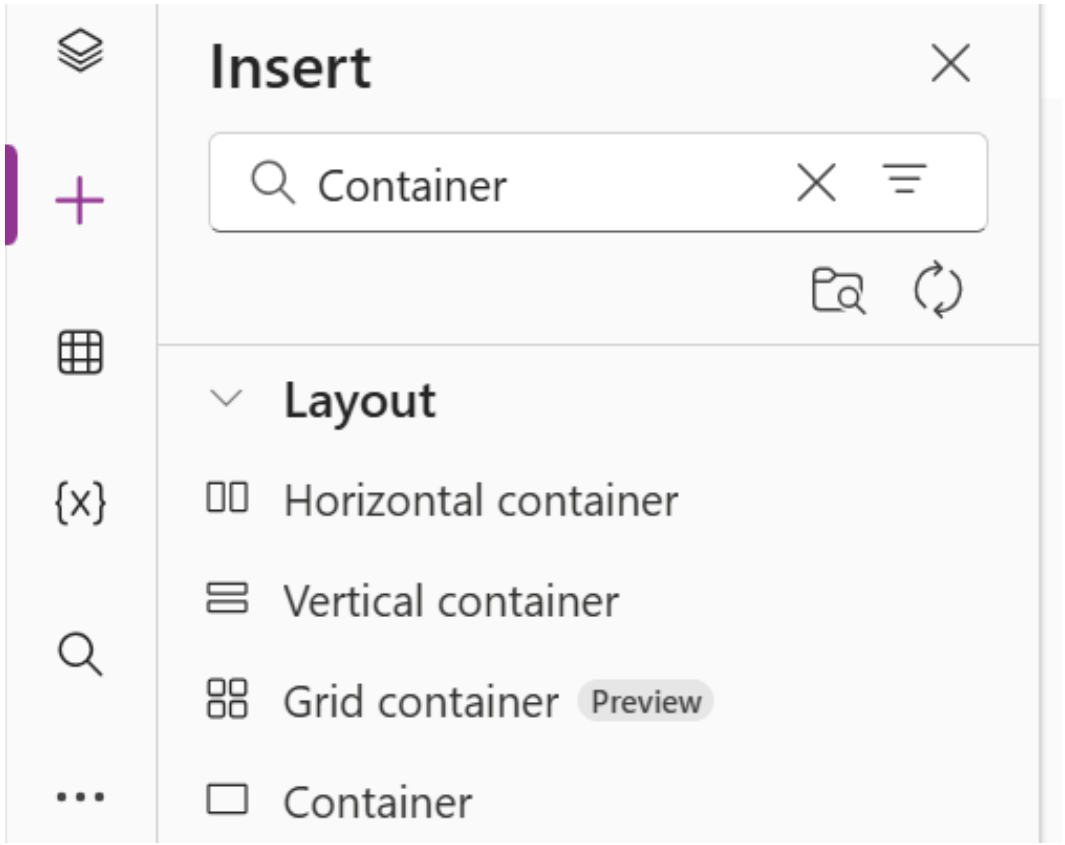
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Part #3: Thriller Bark

Code Blame 11 lines (11 loc) · 318 Bytes

```
With({
    stringDate: Text(Now(), " dd, yyyy @ hh:mm AM/PM"),
    indexMonth: Int(Month(Today())),
    arrayOfMonths: [
        "Jan.","Feb.","Mar.","Apr.",
        "May","Jun.","Jul.","Aug.",
        "Sep.","Oct.","Nov.","Dec."
    ]
},
    Concatenate(Index(arrayOfMonths, indexMonth).Value, stringDate)
)
```

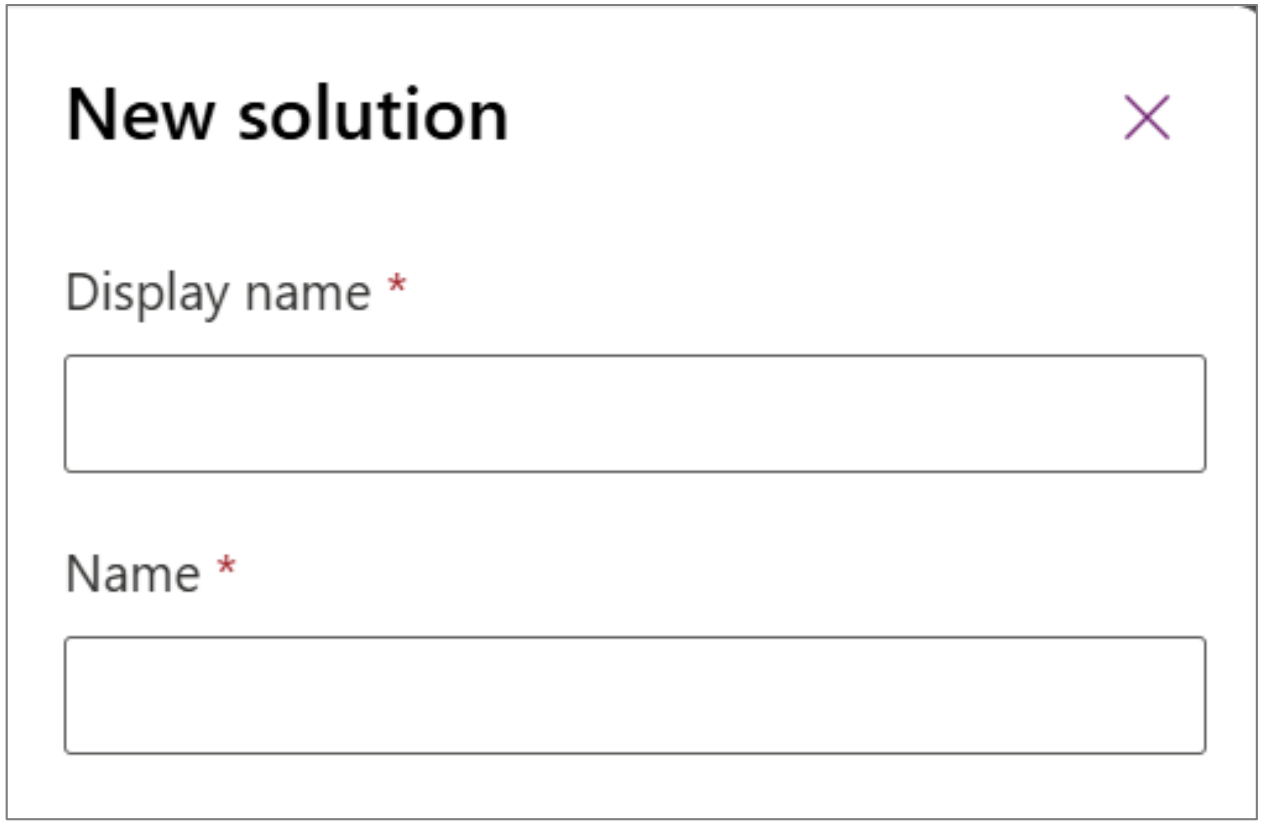
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Part #3: Thriller Bark

```
calcDaysInMonth(
    MM:Number,
    YYYY:Number
):Number = IfError(
    Date(
        Year((Date(YYYY, MM, 1) + 45)),
        Month((Date(YYYY, MM, 1) + 45)),
        1
    ) - Date(YYYY, MM, 1),
    0
);
```

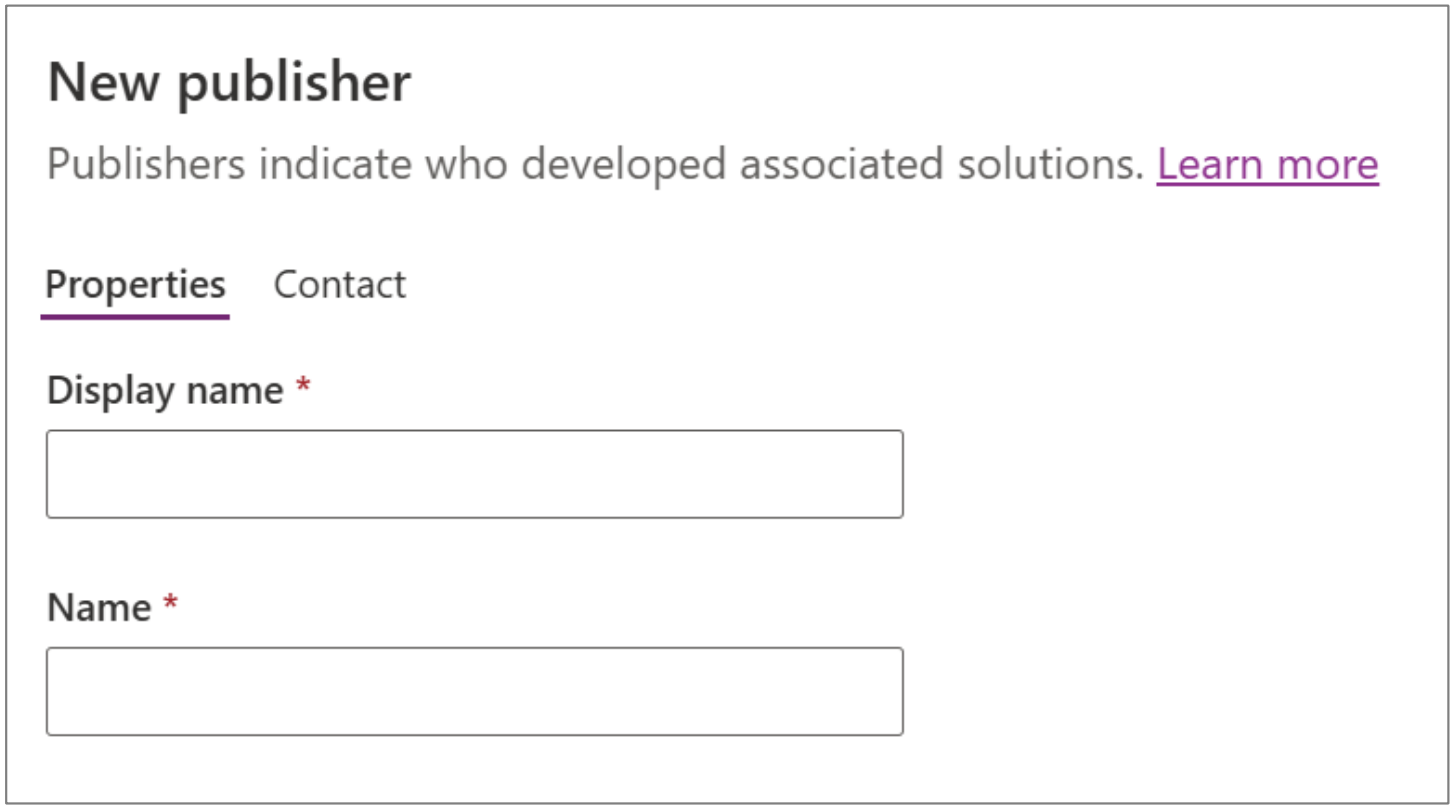
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Part #4: Whole Cake



- ❑ Solutions
- ❑ Publishers
- ❑ Environments
- ❑ Entra ID
- ❑ Service Accounts
- ❑ Component Libraries

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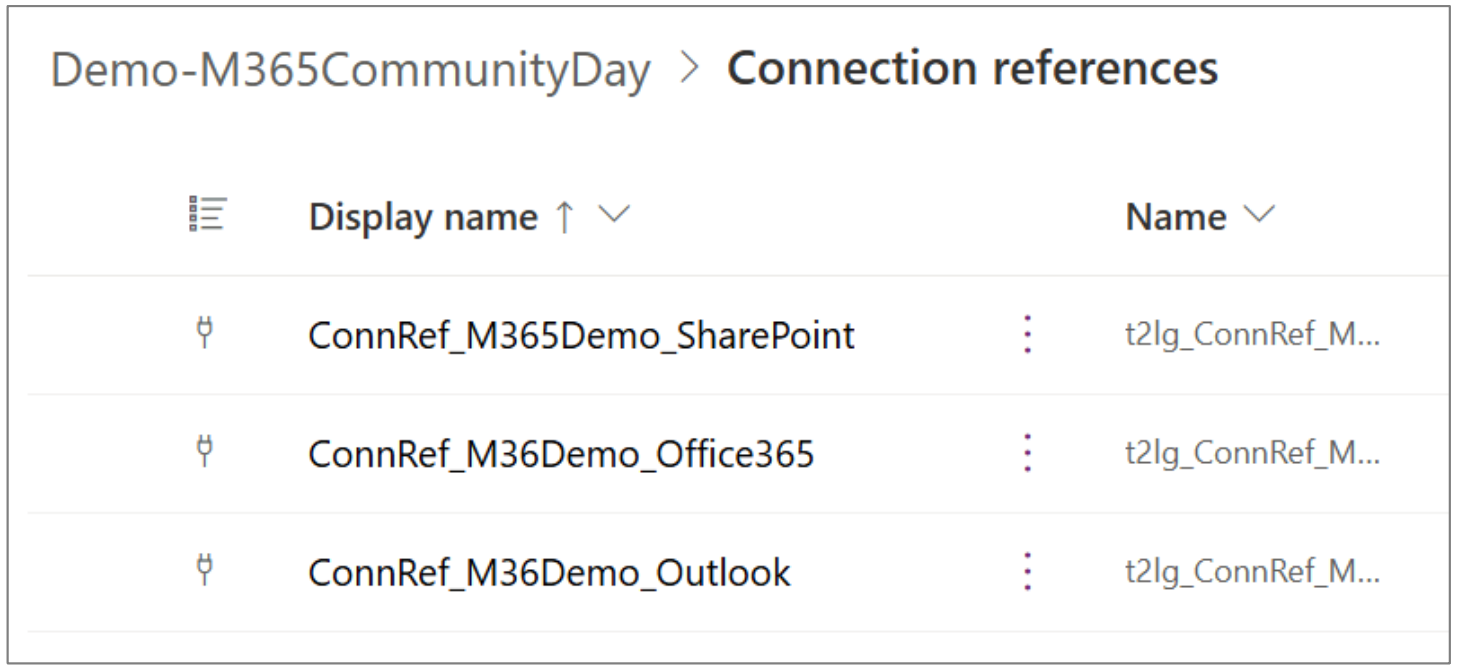
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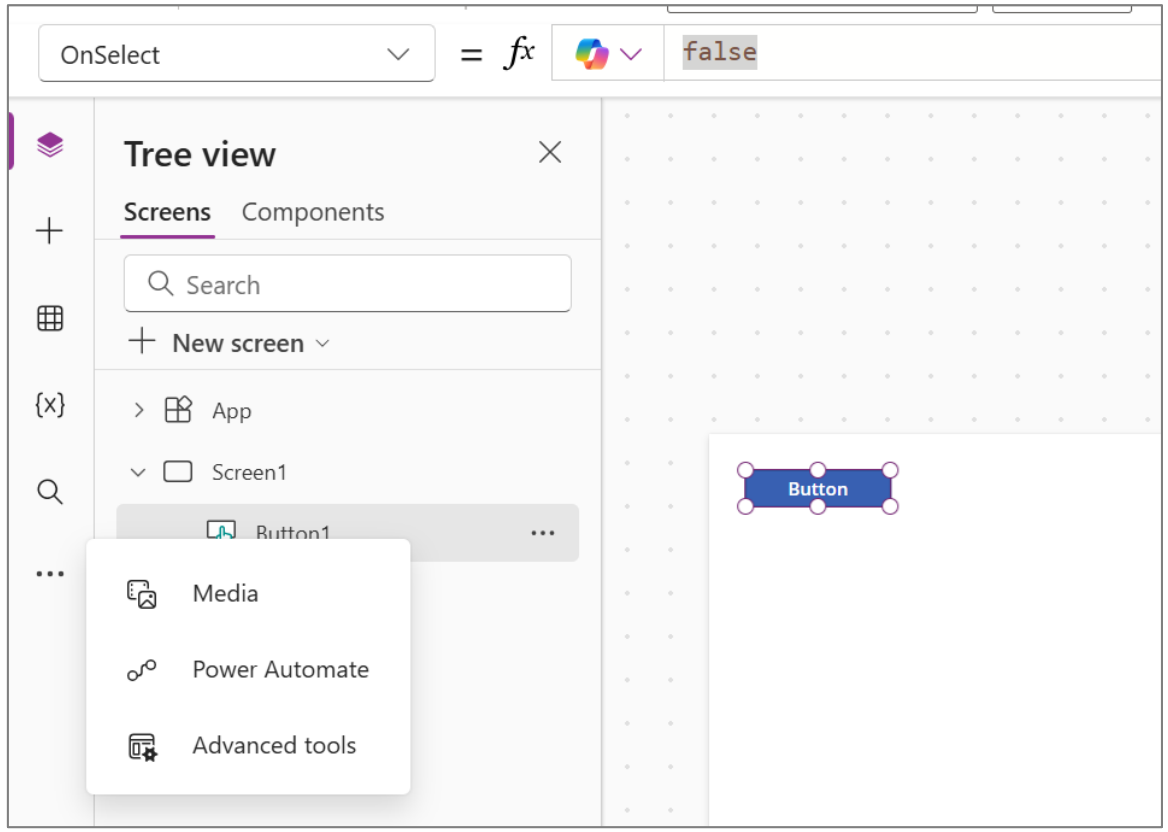
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Part #4: Whole Cake

> **Note**
>
> Component libraries are the recommended way to reuse components across apps. When using a component library, an app maintains dependencies on the components it uses. The app maker will be alerted when the updates to dependent components become available. Hence, all new reusable components should be created within the component libraries instead. An earlier Power Apps feature that allowed importing components from one canvas app to another is retired.

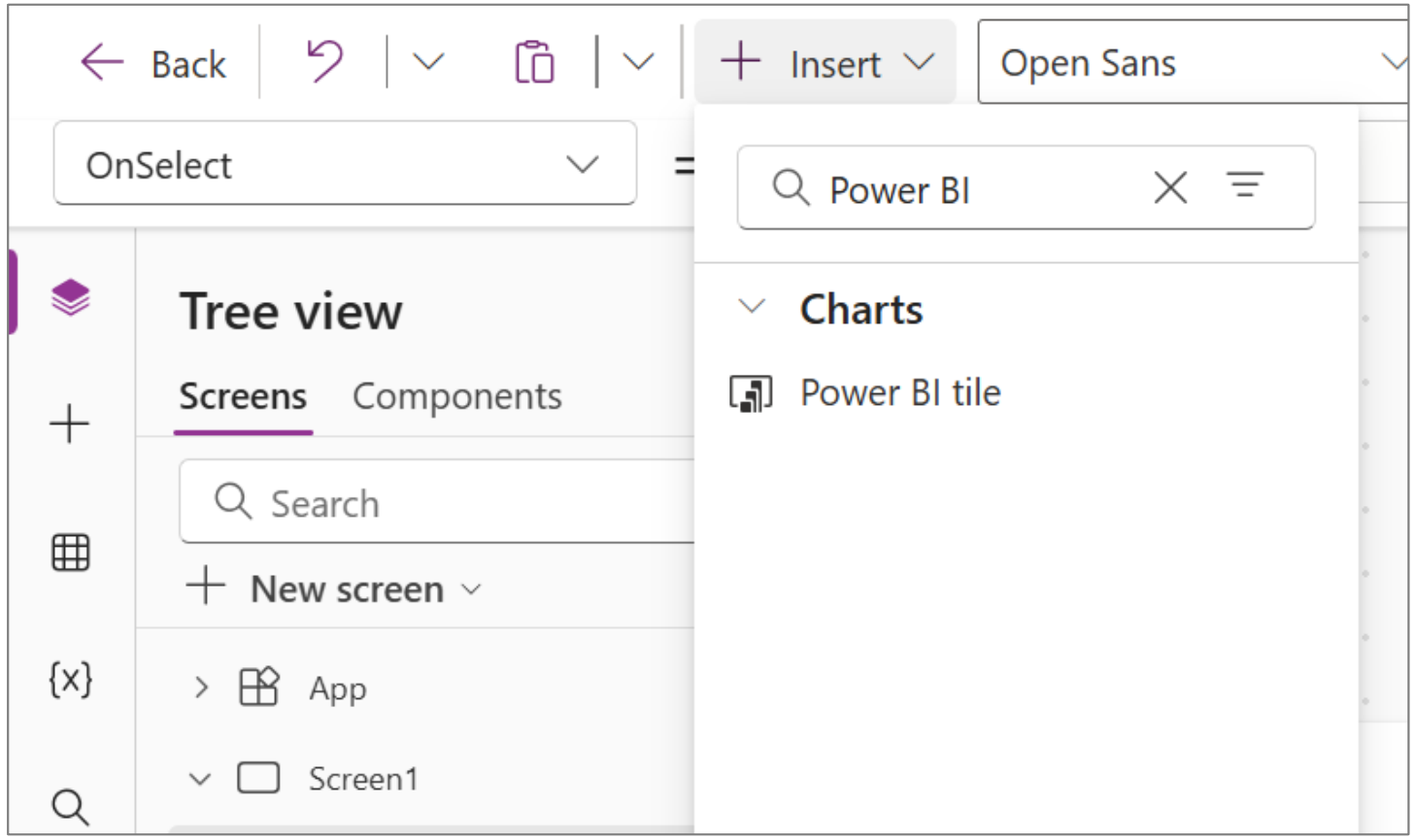
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Part #5: Wano



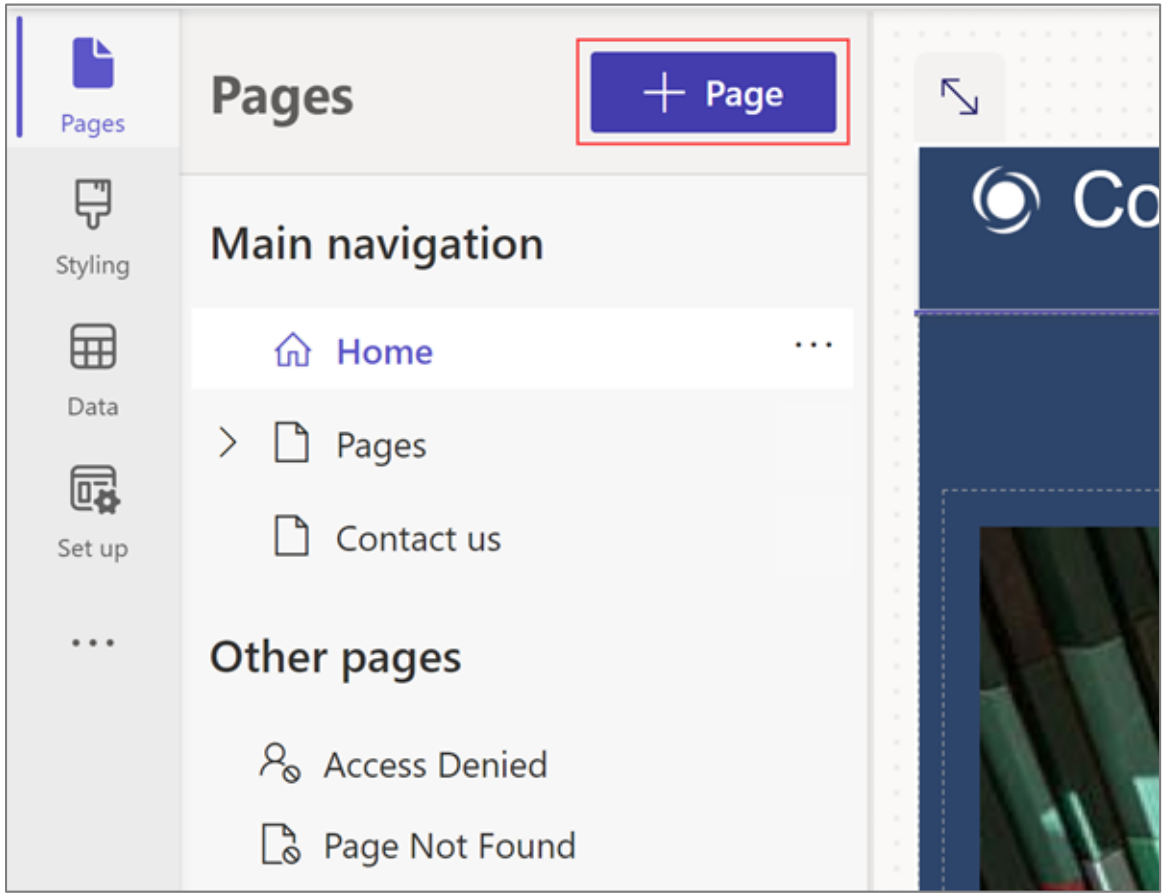
- Power Automate
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- Power BI
  - Dashboards
- Power Pages
  - External Users
- Copilot Studio
  - Channels

Part #5: Wano



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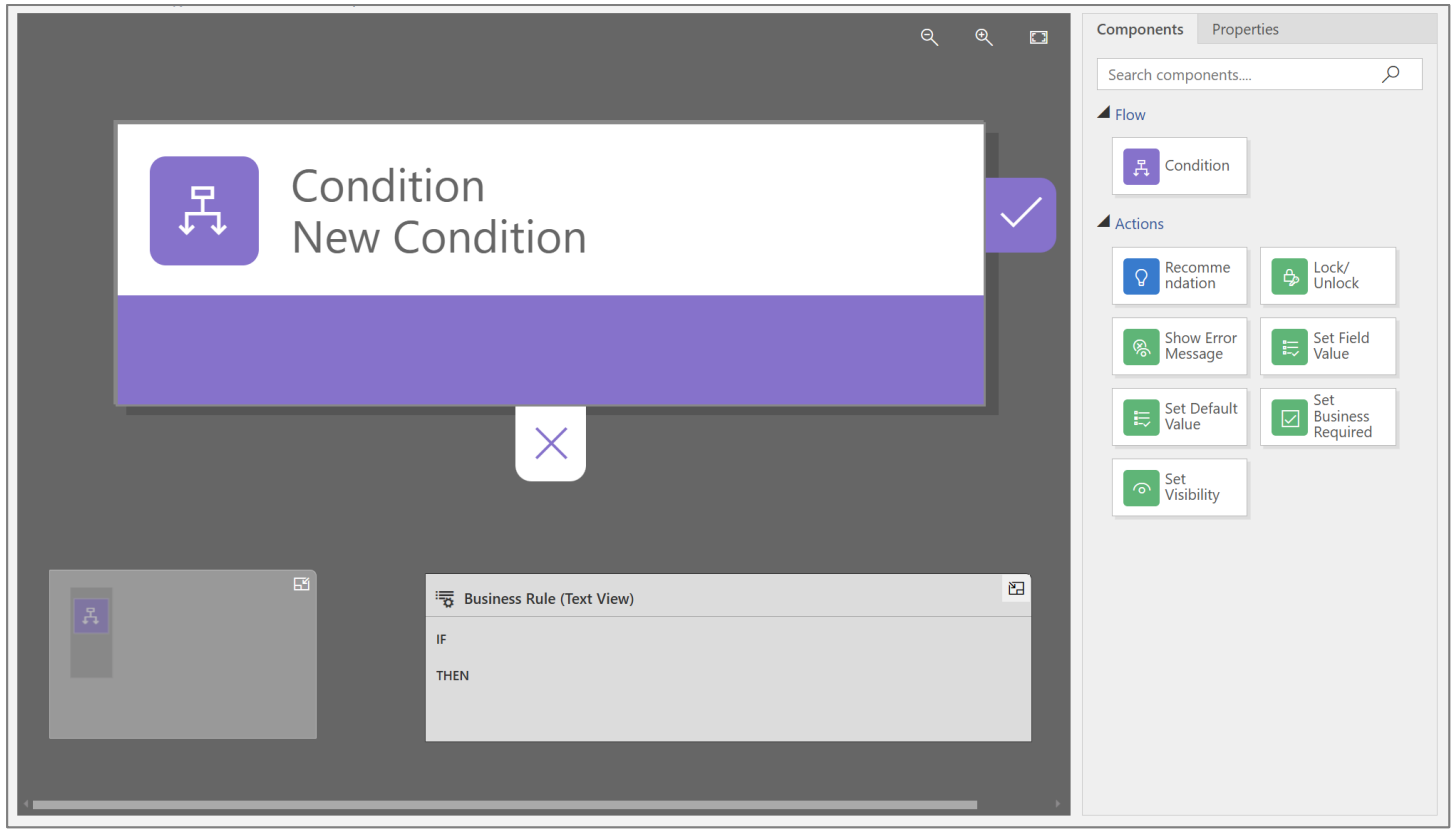
Microsoft channels

Expand table

Channel	Currently available?	Description
Microsoft 365 Copilot	Yes	Make your agent available within Microsoft 365 Copilot's agent list
Microsoft Teams	Yes	Deploy your agent within Teams for chat-based interactions
SharePoint	No	Embed your agent on SharePoint sites for contextual assistance

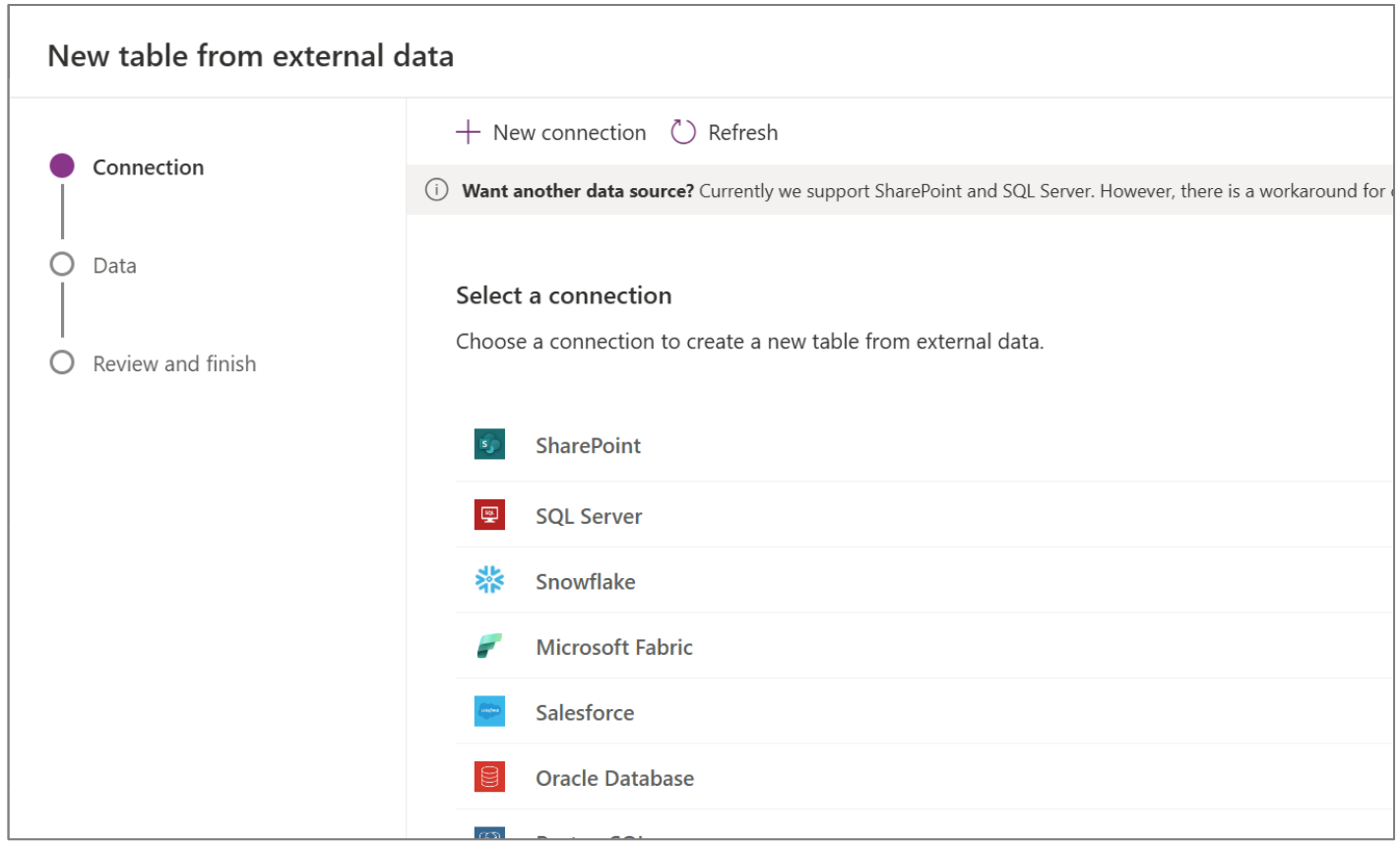
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Part #6: Egghead



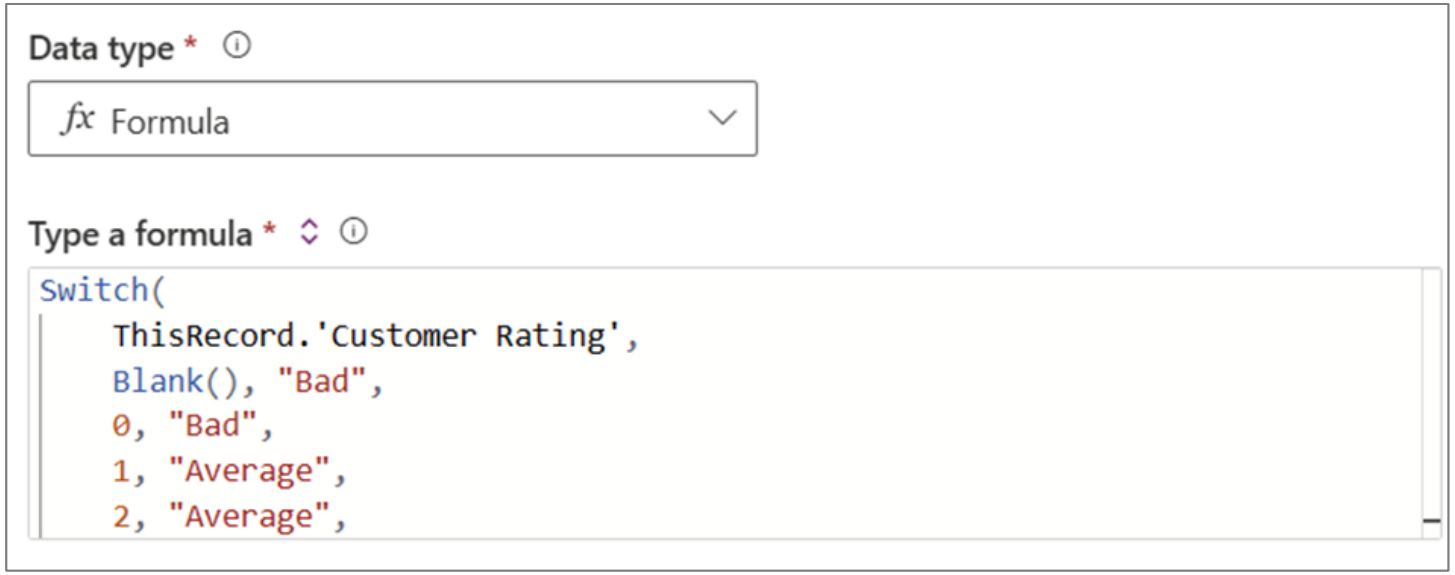
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  - Business Rules
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- DLP
- Pipelines

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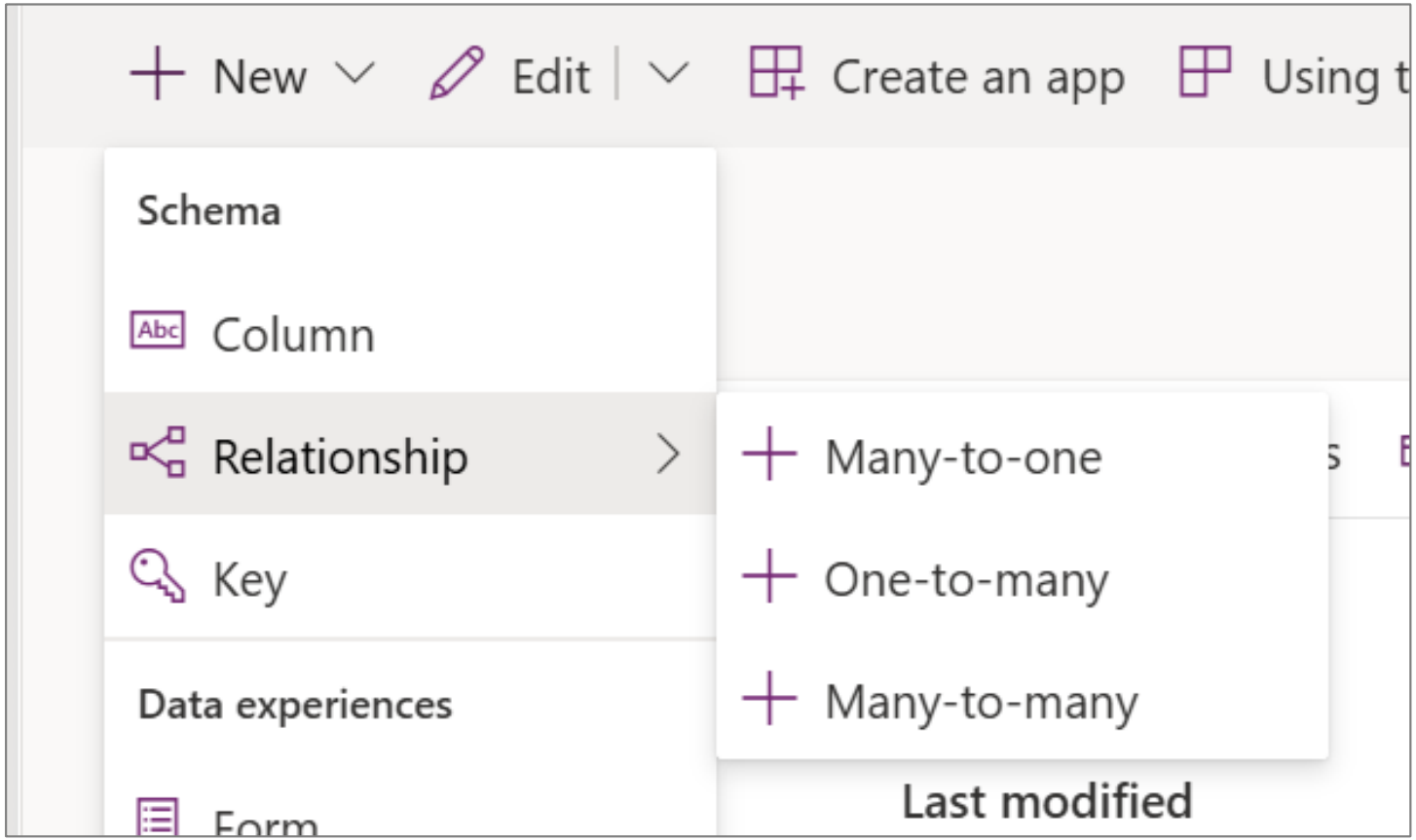
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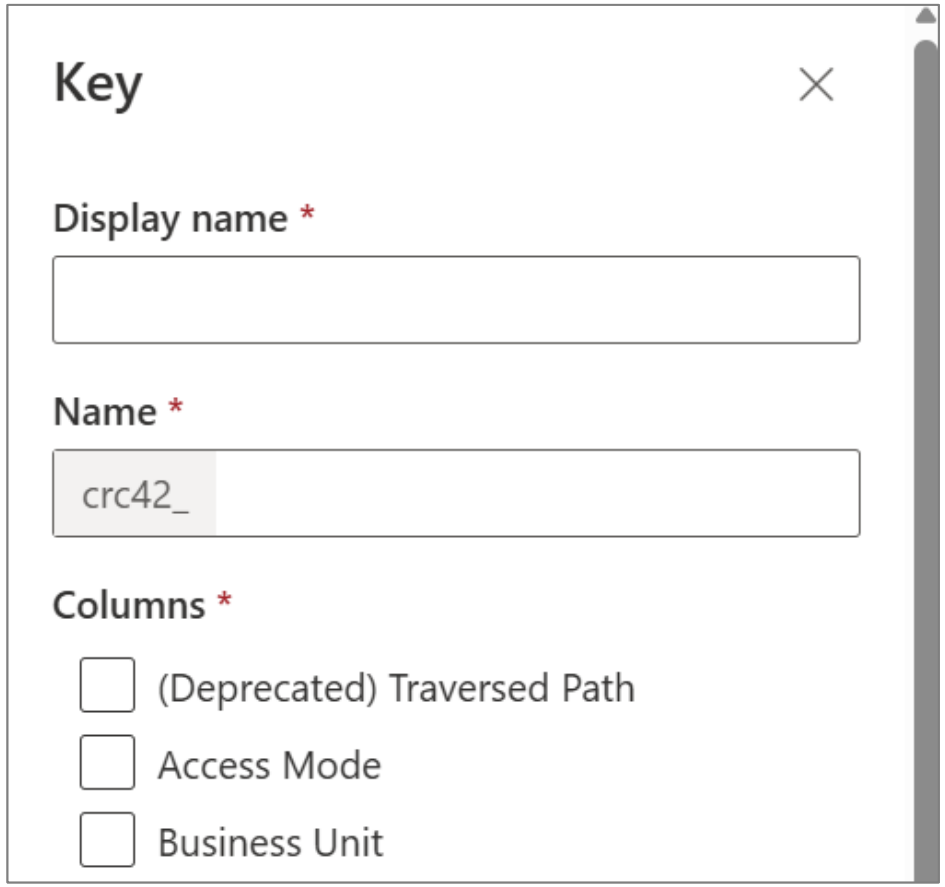
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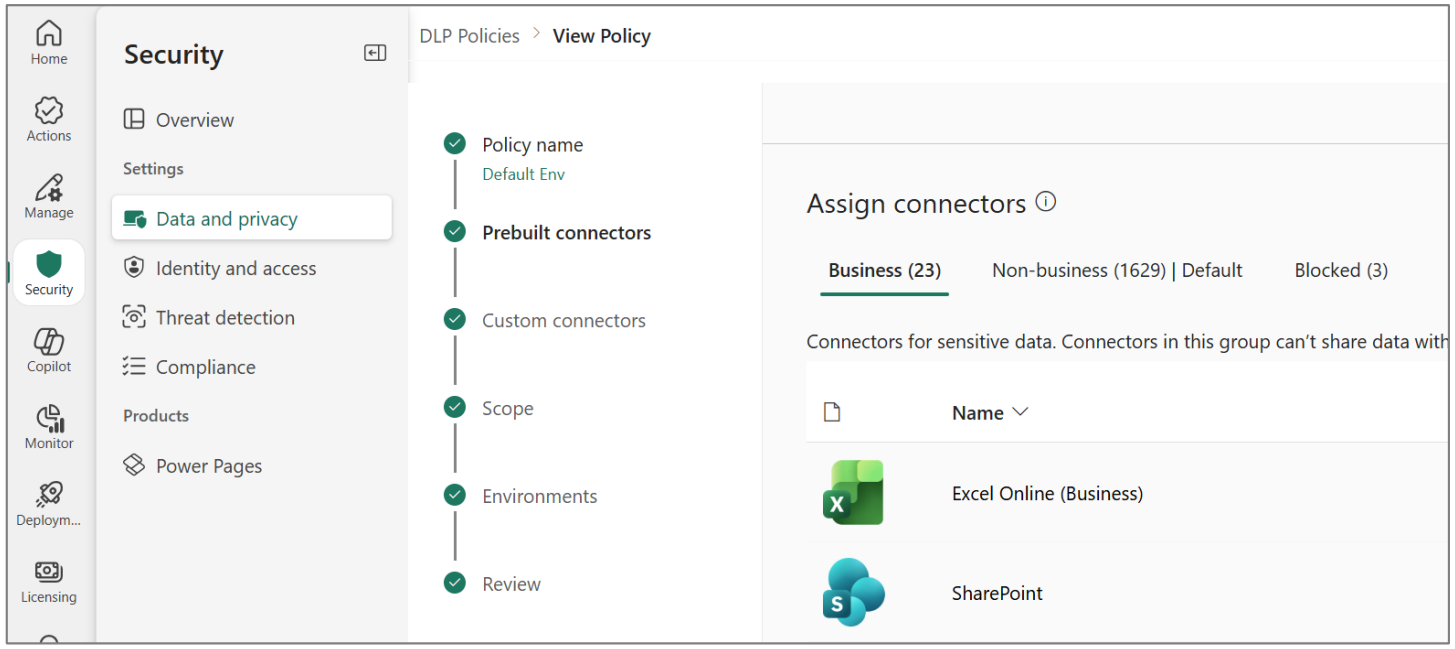
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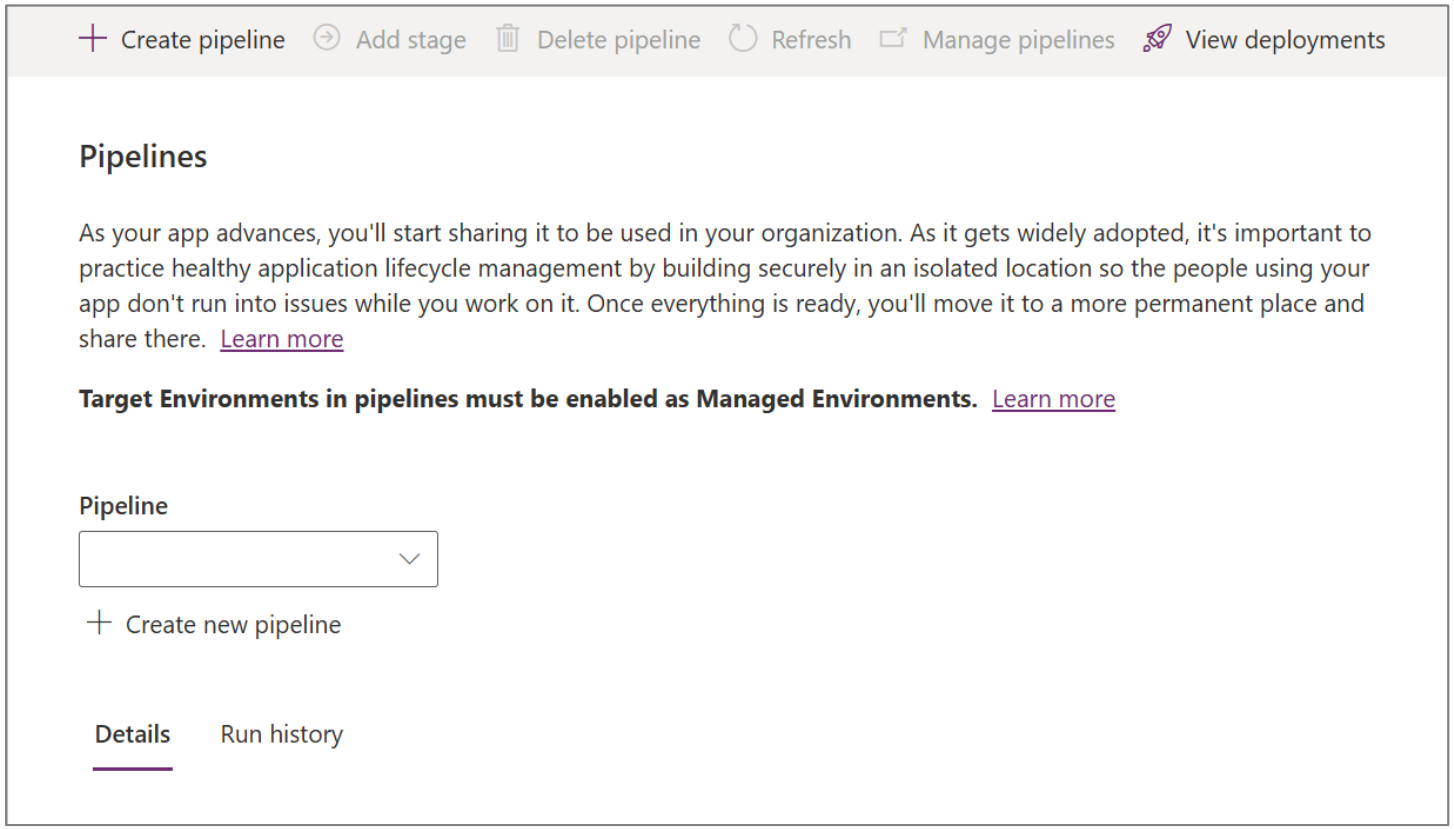
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Recap:

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In this hands-on experience, you'll learn:

- ✅ Power Apps fundamentals – controls, variables, data sources
- ✅ Best practices in UI design and performance
- ✅ How to connect to SharePoint, Dataverse, Excel, and more
- ✅ Working with forms, galleries, and collections
- ✅ Implementing logic, validation, and business rules
- ✅ Creating responsive designs that work on mobile and desktop
- ✅ Publishing and sharing your first business-ready app

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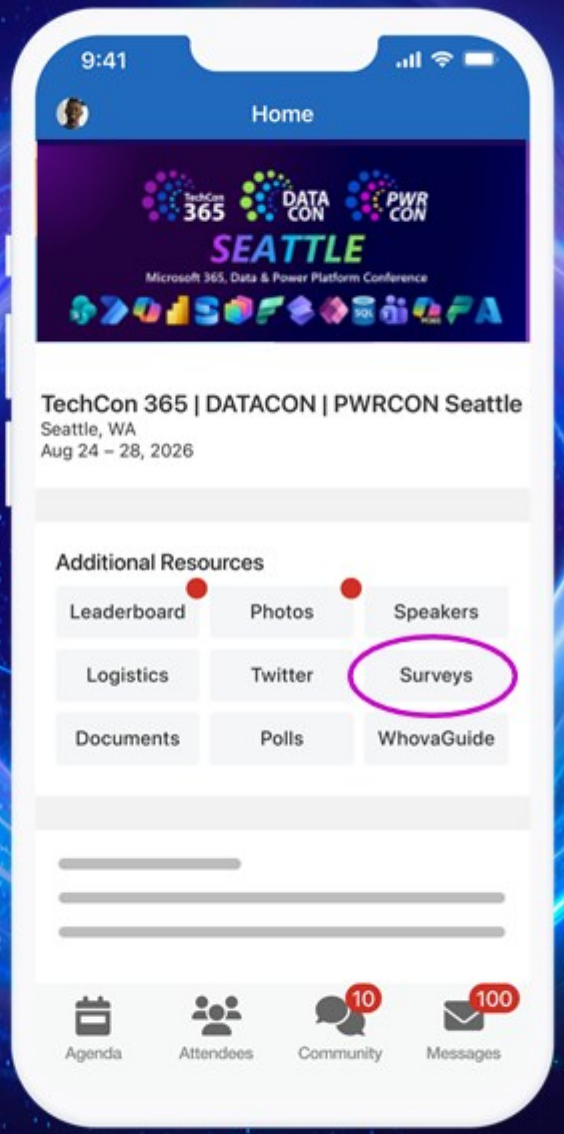
/The2Lakes

@The2Lakes

/losodamus



Charles is a technical speaker and low-code advocate with a pro-code background, specializing in transforming business data architecture. With a passion for innovation, he shares his expertise to solve complex business problems and guide organizations in reimagining their legacy processes, advancing automation initiatives, streamlining operations, and reducing infrastructure costs.



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